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What's common between Mozilla Firefox, VLC Media Player, BitTorrent and Pidgin? Apart from being indispensable applications on any computer, they're unique in belonging to a class of software called 'Open Source'. What this means is that the source code (the set of instructions that enable a program to perform its function) is available for study, modification and redistribution. More commonly abbreviated as FOSS (Free and Open Source Software), this type of software has found widespread acceptance and popularity among communities of software evangelists, who, by critically reviewing and studying the source code of such software, are able to continually improve upon it. (It must be noted here, that 'Free' refers to the freedom to modify and copy, not necessarily meaning free of charge. Most FOSS however, continues to be free of charge as well). This is very similar to a peer reviewed journal, wherein papers are carefully scrutinized by the community before being published. Today, the term covers a massive range of software, from operating systems (GNU/Linux, FreeBSD, Android) and servers (Apache, Moodle, Wordpress) to popular applications and even programming languages like PHP and Python.

The Indian story

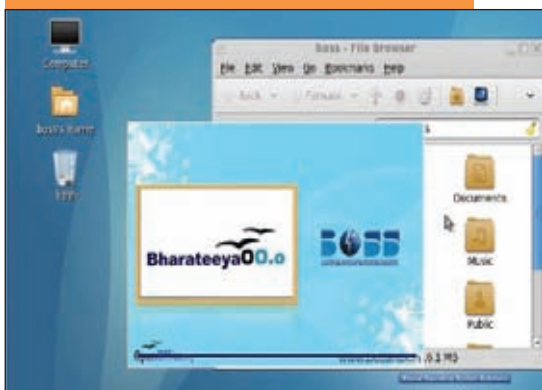
The ideology of Open Source holds a lot of promise for India and has



open
source
in india

How FOSS is transforming our society by taking computing to the masses

found favour in a number of projects throughout the country. The Department of Information Technology (DIT), Government of India, has held a keen interest in FOSS, leading to the formation of the NRCFOSS (National Resource Centre for Free and Open Source Software) in 2005. NRCFOSS serves as the nodal authority, co-ordinating various FOSS projects in India. According to its web site <http://www.nrcfoss.org.in/>, it was formed "with the twin roles of bridging the digital divide as well as strengthening the Indian software industry". The implementing agencies include C-DAC (Centre for Development of Advanced Computing), Anna University (AU-KBC), IIT Bombay and IIT Madras. Some of its key achievements include the development of the BOSS Linux operating system, the setting up of the GNU Compiler Collection (GCC) Resource Centre at IIT Bombay and starting of the Centre of Excellence for mobile internet devices jointly at C-DAC Chennai and DIT.



Open source story in India

One of the primary objectives behind the formulation of the NRC-FOSS was increasing computer literacy by imparting computer education to students as well as teachers throughout the country. It was soon realized, that one major hurdle in taking computing to the masses was the lack of computer software in native Indian languages. Since only 11 per cent of our population can speak English, the urgent need for developing tools and products that are highly customised to Indian needs was realised.

One of the earliest efforts in this direction was the IndLinux project, an independent venture, jointly started by two Open Source enthusiasts, Prakash Advani and Venkatesh Hariharan. The project's aim was to create a Linux distribution which would provide full support to all major Indian languages. This included the development of fonts, translators as well as changes to the kernel, which would ultimately "bring the benefits of Information Technology down to the Indian masses." (<http://indlinux.org>) Karunakar Guntupalli, a key developer of the IndLinux project, stated that "It was the first Linux distribution to have all Indian languages supported by default, and was mainly released as a live CD which could also be installed. It was derived from another live CD called Morphix. The main goal was to demonstrate Indian language support in Linux. A Live CD was the most effective and non-intrusive way, since the language support hadn't yet reached the mainstream level. The IndLinux Rangoli base was also used in the Nepalese (Nepalinux) and Bhutanese (DzongkhaLinux) language versions"

Development on the IndLinux Rangoli distribution stopped in a couple of years, once mainstream Linux distributions started including Indic languages, and a user could choose an Indic language during the installation process itself.

Meanwhile, C-DAC Chennai set about making an operating system that would suit Indian requirements by harnessing the power of Open Source. This resulted in BOSS (Bharat